

Project Proposal

Project Type: Smart India Hackathon (SIH) Problem Statement - SIH 1683

Team Details

Team Members Count: 3

Team Member	Student ID	Name	Phone Number
Team Member 1	24CS080	Varshil Patel	8200468354
Team Member 2	24CS074	Rudra Patel	7041802201
Team Member 3	24CS078	Trusha Patel	8320852022

Title

Development of an AI/ML-Based Explainable System for Detection of Face-Swap Deepfake Videos

Introduction

With the rapid growth of Artificial Intelligence (AI) and Generative AI, creating realistic deepfake videos has become easier than ever. Face-swap deepfakes, where one person's face is replaced with another's, are increasingly being used for misinformation, identity theft, financial fraud, cybercrime, political manipulation, and the creation of non-consensual content. As these fake videos become more realistic, distinguishing them from genuine ones has become a major challenge for law enforcement agencies, digital forensic experts, and cybersecurity organizations. This project aims to address this challenge by developing an intelligent and explainable AI-based system that can accurately detect face-swap deepfake videos and provide supporting forensic evidence.

Objective

The primary objective of this project is to develop an AI/ML-based hybrid deepfake detection system capable of identifying face-swap manipulated videos. Instead of providing only a **Real** or **Fake** prediction, the system will generate a detailed forensic report explaining the reasons behind the decision, including suspicious regions, detected abnormalities, confidence scores, and the likely deepfake generation technique.

Proposed Methodology

The proposed solution follows a multi-stage detection pipeline. The uploaded video is first processed using **OpenCV** and **FFmpeg** to extract video frames and audio. **RetinaFace** is then used to accurately detect and crop facial regions. Visual manipulation artifacts are analyzed using deep learning models such as **EfficientNet**, while temporal inconsistencies across consecutive frames are identified using **BiLSTM**. To

uncover hidden manipulation patterns, **Fast Fourier Transform (FFT)** is applied for frequency-domain analysis. Audio features are extracted using **Librosa**, and **SyncNet** is used to verify whether lip movements are synchronized with the spoken audio. Finally, all extracted features are combined to make the final prediction. Explainable AI techniques such as **Grad-CAM** and **SHAP** are incorporated to highlight manipulated regions and provide visual evidence supporting the system's decision.

Expected Outcome

The proposed system will:

- Detect face-swap deepfake videos with high accuracy.
- Analyze spatial, temporal, frequency, and audio-visual inconsistencies.
- Generate an explainable forensic report with confidence scores and frame-level evidence.
- Assist digital forensic experts, cybersecurity professionals, and security agencies in verifying the authenticity of digital videos.

Technology Stack

- **Programming Language:** Python
- **Computer Vision:** OpenCV, RetinaFace
- **Deep Learning:** PyTorch, EfficientNet, BiLSTM
- **Frequency Analysis:** Fast Fourier Transform (FFT)
- **Audio Processing:** Librosa, SyncNet
- **Explainable AI:** Grad-CAM, SHAP
- **Backend:** FastAPI
- **Frontend:** React
- **Deployment:** Docker

Conclusion

This project proposes a practical, explainable, and scalable AI/ML solution for detecting face-swap deepfake videos. By combining spatial, temporal, frequency, and audio-visual analysis, the system aims to achieve high detection accuracy while providing clear forensic evidence to support its decisions.